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SUB NAME: FUNDEMENTALS OF DATA SCIENCE

Case Study 1 — Employee Salaries

Given the salaries (in ₹ lakhs):

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Mean, Median, and Mode

Mean

[
\text{Mean}=\frac{4+5+5+6+7+8+8+9+30}{9}
=\frac{82}{9}
=9.11
]

Answer: Mean = 9.11 lakhs

Median

There are 9 values, so the median is the middle (5th) value.

Ordered data:

4, 5, 5, 6, 7, 8, 8, 9, 30

The **5th value = 7**

Answer: Median = 7 lakhs

Mode

The numbers **5** and **8** each occur twice.

Answer: Mode = 5 lakhs and 8 lakhs (Bimodal)

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2. Range

[
\text{Range}=\text{Maximum}-\text{Minimum}
=30-4=26
]

Answer: Range = 26 lakhs

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3. Variance and Standard Deviation (Population)

Mean = **9.11**

Salary (x)	(x-\bar{x})	((x-\bar{x})^2)
4	−5.11	26.12
5	−4.11	16.90
5	−4.11	16.90
6	−3.11	9.68
7	−2.11	4.46
8	−1.11	1.23
8	−1.11	1.23
9	−0.11	0.01
30	20.89	436.23

Sum of squared deviations:

[
\sum (x-\bar{x})^2=512.76
]

Variance

[
\sigma^2=\frac{512.76}{9}=56.97
]

Standard Deviation

[
\sigma=\sqrt{56.97}\approx7.55
]

Answers:

- **Variance = 56.97**
- **Standard Deviation = 7.55 lakhs**

(If your class uses the sample formula, then variance = 64.10 and standard deviation ≈ 8.01.)

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4. Identify the Outlier

The salary **30 lakhs** is much larger than the other salaries (which range from 4–9 lakhs).

Answer: 30 lakhs is the outlier.

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5. Which Measure Best Represents the Typical Salary?

Median (7 lakhs) is the best measure.

Reason:

- The outlier (30 lakhs) increases the **mean** to **9.11 lakhs**, making it misleading.
- The **median** is not affected by the outlier and better represents the typical employee salary.

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2) Case Study 2 — Website Response Time

Given data (in milliseconds):

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the Mean and Median

Mean

$$
\text{Mean}=\frac{120+125+128+130+132+135+140+145+150+300}{10}
$$

$$
=\frac{1505}{10}=150.5\text{ ms}
$$

Mean = 150.5 ms

Median

There are 10 values (even number), so the median is the average of the 5th and 6th values.

5th value = 132

6th value = 135

$$
\text{Median}=\frac{132+135}{2}=133.5\text{ ms}
$$

Median = 133.5 ms

---

2. Find the Range

[
\text{Range}=\text{Maximum}-\text{Minimum}
]

[
=300-120=180\text{ ms}
]

Range = 180 ms

---

3. Calculate the Variance and Standard Deviation

Using the population formula:

Value (x)	(x-\bar{x})	((x-\bar{x})^2)
120	−30.5	930.25
125	−25.5	650.25
128	−22.5	506.25
130	−20.5	420.25
132	−18.5	342.25
135	−15.5	240.25
140	−10.5	110.25
145	−5.5	30.25
150	−0.5	0.25

Value (x)	(x-\bar{x})	((x-\bar{x})^2)
300	149.5	22350.25

Sum of squared deviations:

[
25581.5
]

Variance

[
\sigma^2=\frac{25581.5}{10}=2558.15
]

Variance = 2558.15 ms^2

Standard Deviation

[
\sigma=\sqrt{2558.15}\approx50.58
]

Standard Deviation ≈ 50.58 ms

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4. Determine Whether the Data is Right-Skewed or Left-Skewed

- **Mean = 150.5 ms**
- **Median = 133.5 ms**

- One very large value (300 ms) pulls the mean upward.

Answer: The data is right-skewed (positively skewed).

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5. Why is the Median More Useful Than the Mean?

The response time of 300 ms is an outlier.

- The mean (150.5 ms) is increased by this unusually high value.
- The median (133.5 ms) is not greatly affected by the outlier and better represents the typical website response time.

Answer: The median is more useful because it is less affected by extreme values (outliers) and gives a better measure of the typical response time.

Case Study 3 — Student Performance

Given data (already in ascending order):

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

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1. Find Q1, Q2, and Q3

There are 12 observations (even number).

Q2 (Median)

Median is the average of the 6th and 7th values.

6th value = 52

7th value = 55

$$
Q_2=\frac{52+55}{2}=53.5
$$

Q2 = 53.5

Q1

First half:

42, 45, 48, 50, 52, 52

Median of first half:

$$
Q_1=\frac{48+50}{2}=49
$$

Q1 = 49

Q3

Second half:

55, 58, 60, 62, 65, 95

Median of second half:

[
Q_3=\frac{60+62}{2}=61
]

Q3 = 61

---

2. Calculate the IQR

[
IQR=Q_3-Q_1
]

[
=61-49=12
]

IQR = 12

---

3. Use the 1.5 × IQR Rule to Identify Outliers

Lower Limit

[
Q_1-1.5(IQR)=49-1.5(12)
]

[
=49-18=31
]

Upper Limit

$$Q_3+1.5(IQR)=61+18=79$$

Values below 31 or above 79 are outliers.

Only 95 is greater than 79.

Outlier = 95

---

4. Find the Mean and Median

Mean

$$\text{Sum}=42+45+48+50+52+52+55+58+60+62+65+95=684$$

$$\text{Mean}=\frac{684}{12}=57$$

Mean = 57

Median

Already calculated:

Median = 53.5

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5. Explain How the Outlier Affects the Mean

The value 95 is much higher than the other marks.

- **It increases the mean from what would otherwise be closer to the center of the data.**
- **The median (53.5) is not greatly affected because it depends only on the middle values.**

Answer: The outlier 95 pulls the mean upward, making it larger than the median. Therefore, the median better represents the typical student performance.

Case Study 4 — E-Commerce Orders

Given data (number of orders over 10 days):

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

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1. Calculate the Mean, Median, and Mode

Mean

$$
\text{Mean}=\frac{18+20+22+22+24+25+26+28+30+35}{10}
$$

$$
=\frac{250}{10}=25
$$

Mean = 25

Median

There are 10 values, so the median is the average of the 5th and 6th values.

5th value = 24

6th value = 25

$$
\text{Median}=\frac{24+25}{2}=24.5
$$

Median = 24.5

Mode

The value 22 occurs twice, while all others occur once.

Mode = 22

---

2. Find the Variance and Standard Deviation

Using the population formula:

x	$x - 25$	$(x - 25)^2$
18	−7	49
20	−5	25
22	−3	9
22	−3	9
24	−1	1
25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

[
\sum (x-\bar{x})^2 = 228
]

Variance

[
\sigma^2=\frac{228}{10}=22.8
]

Variance = 22.8

Standard Deviation

$$
\sigma=\sqrt{22.8}\approx4.77
$$

Standard Deviation ≈ 4.77

---

3. Calculate the Coefficient of Variation (CV)

$$
CV=\frac{\text{Standard Deviation}}{\text{Mean}}\times100
$$

$$
=\frac{4.77}{25}\times100
$$

$$
=19.08\%
$$

Coefficient of Variation ≈ 19.08%

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4. What Does the Standard Deviation Tell the Company?

The standard deviation (4.77) indicates how much the daily orders vary from the average.

- A small standard deviation means the daily orders are fairly consistent.

- **A larger standard deviation means there is more variation in daily orders.**

Answer: Since the standard deviation is about 4.77 orders, the daily orders are fairly close to the average of 25 orders.

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5. Is the Order Data Highly or Moderately Variable?

The Coefficient of Variation (19.08%) is below 20%, indicating relatively low variability.

Answer: The order data is moderately variable (fairly consistent), not highly variable.

Case Study 5 — Comparing Two Data Science Models

Given prediction errors:

- **Model A: 2, 3, 3, 4, 4, 5, 5, 6**
- **Model B: 1, 1, 2, 4, 5, 7, 8, 9**

There are 8 observations in each model.

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1. Calculate the Mean Error

Model A

[
\text{Mean}=\frac{2+3+3+4+4+5+5+6}{8}
=\frac{32}{8}=4
]

Mean (A) = 4

Model B

[
\text{Mean}=\frac{1+1+2+4+5+7+8+9}{8}
=\frac{37}{8}=4.625
]

Mean (B) = 4.625

---

2. Calculate the Variance (Population)

Model A

x	$x-4$	$(x-4)^2$
2	−2	4
3	−1	1
3	−1	1
4	0	0
4	0	0
5	1	1

x	$x-4$	$(x-4)^2$
5	1	1
6	2	4

$$\sum (x-\bar{x})^2 = 12$$

$$\text{Variance}=\frac{12}{8}=1.5$$

Variance (A) = 1.5

Model B

Mean = 4.625

x	$x-4.625$	$(x-4.625)^2$
1	−3.625	13.1406
1	−3.625	13.1406
2	−2.625	6.8906
4	−0.625	0.3906
5	0.375	0.1406
7	2.375	5.6406

x	$x - 4.625$	$(x - 4.625)^2$
8	3.375	11.3906
9	4.375	19.1406

[
\sum (x-\bar{x})^2 = 69.875
]

[
\text{Variance}=\frac{69.875}{8}=8.7344
]

Variance (B) ≈ 8.73

---

3. Calculate the Standard Deviation

Model A

[
SD=\sqrt{1.5}\approx1.22
]

Standard Deviation (A) ≈ 1.22

Model B

[
SD=\sqrt{8.7344}\approx2.96
]

Standard Deviation (B) ≈ 2.96

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4. Which Model Has More Consistent Predictions?

A smaller standard deviation means the prediction errors are more consistent.

- **Model A: SD ≈ 1.22**
- **Model B: SD ≈ 2.96**

Answer: Model A has more consistent predictions because its prediction errors vary less.

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5. Can the Model with the Lower Mean Error Necessarily Be Considered Better?

No.

A lower mean error is important, but it does not always mean the model is better. Other factors should also be considered, such as:

- **Variance or standard deviation (consistency)**
- **Accuracy**
- **Precision and recall**
- **F1-score**
- **Performance on unseen (test) data**

A model with a slightly lower mean error but very high variability may perform less reliably than a model with slightly higher mean error but much more consistent predictions.
